============================================================ HRS STANDARD LIBRARY STEP-BY-STEP IMPLEMENTATION GUIDE ============================================================ Objective --------- Create your own custom C library named "HRS Standard Library" and use it exactly like a standard library. Example: #include instead of #include "hrs.h"

============================================================ STEP 1 : CREATE PROJECT FOLDER ============================================================ Open Command Prompt. Command: mkdir HRS_Library cd HRS_Library

============================================================ STEP 2 : CREATE THE PROJECT STRUCTURE ============================================================ Command: mkdir include mkdir src mkdir examples mkdir bin mkdir lib mkdir docs Verify: tree /f Expected Structure HRS_Library │ ├── include ├── src ├── examples ├── bin ├── lib └── docs

============================================================ STEP 3 : CREATE THE FILES ============================================================ Command: type nul > include\hrs.h type nul > src\hrs.c type nul > examples\demo.c type nul > README.md Verify: tree /f Expected Structure HRS_Library │ README.md │ ├── include │ hrs.h │ ├── src │ hrs.c │ ├── examples │ demo.c │ ├── bin ├── lib └── docs

============================================================ STEP 4 : OPEN PROJECT IN VS CODE ============================================================ Command: code . ============================================================ STEP 5 : WRITE THE HEADER FILE ============================================================ File: include\hrs.h Code: ------------------------------------------------------------ #ifndef HRS_H #define HRS_H /* HRS Standard Library Version 1.0 */ void hrs_banner(); void hrs_line(); void hrs_version(); void hrs_success(const char *message); int hrs_add(int a, int b); #endif ------------------------------------------------------------

============================================================ STEP 6 : WRITE THE SOURCE FILE ============================================================ File: src\hrs.c Code: ------------------------------------------------------------ #include #include "../include/hrs.h" void hrs_banner() { printf("============================================================\n"); printf(" HRS STANDARD LIBRARY v1.0\n"); printf(" Developed by Himanshu Rajak\n"); printf("============================================================\n"); } void hrs_line() { printf("------------------------------------------------------------\n"); } void hrs_version() { printf("Library Version : 1.0\n"); } void hrs_success(const char *message) { printf("[SUCCESS] %s\n", message); } int hrs_add(int a, int b) { return a + b; } ------------------------------------------------------------ ============================================================ STEP 7 : WRITE THE DEMO PROGRAM ============================================================ File: examples\demo.c Code: ------------------------------------------------------------ #include #include "hrs.h" int main() { hrs_banner(); hrs_version(); hrs_line(); printf("10 + 20 = %d\n", hrs_add(10,20)); hrs_success("Library Loaded Successfully"); return 0; } ------------------------------------------------------------ ============================================================ STEP 8 : COMPILE THE PROJECT ============================================================ Command: gcc examples\demo.c src\hrs.c -Iinclude -o bin\demo.exe Explanation examples\demo.c Main program src\hrs.c Library implementation -Iinclude Search header files inside the include folder -o bin\demo.exe Generate executable in the bin folder

============================================================ STEP 9 : RUN THE PROGRAM ============================================================ Command: bin\demo.exe Expected Output

==================================================== HRS STANDARD LIBRARY v1.0 Developed by Himanshu Rajak ==================================================== Library Version : 1.0 -------------------------------------------------- 10 + 20 = 30 [SUCCESS] Library Loaded Successfully ============================================================ STEP 10 : FIND GCC INSTALLATION ============================================================ Command: where gcc Example Output C:\MinGW\bin\gcc.exe This means GCC is installed in C:\MinGW ============================================================ STEP 11 : COPY hrs.h TO THE GCC INCLUDE DIRECTORY ============================================================ Command: copy include\hrs.h C:\MinGW\include\ Expected Output 1 file(s) copied. Verify dir C:\MinGW\include\hrs.h ============================================================ STEP 12 : MODIFY THE DEMO PROGRAM ============================================================ Replace #include "hrs.h" with #include Now the file becomes ------------------------------------------------------------ #include #include int main() { hrs_banner(); hrs_version(); hrs_line(); printf("10 + 20 = %d\n", hrs_add(10,20)); hrs_success("Library Loaded Successfully"); return 0; } ------------------------------------------------------------ ============================================================ STEP 13 : COMPILE AGAIN ============================================================ Since hrs.h is now installed in GCC's include directory, -Iinclude is no longer required. Command gcc examples\demo.c src\hrs.c -o bin\demo.exe ============================================================ STEP 14 : RUN THE PROGRAM ============================================================ Command bin\demo.exe Expected Output ==================================================== HRS STANDARD LIBRARY v1.0 Developed by Himanshu Rajak ==================================================== Library Version : 1.0 -------------------------------------------------- 10 + 20 = 30 [SUCCESS] Library Loaded Successfully ============================================================ PROJECT STRUCTURE ============================================================ HRS_Library │ ├── include │ └── hrs.h │ ├── src │ └── hrs.c │ ├── examples │ └── demo.c │ ├── bin ├── lib ├── docs └── README.md